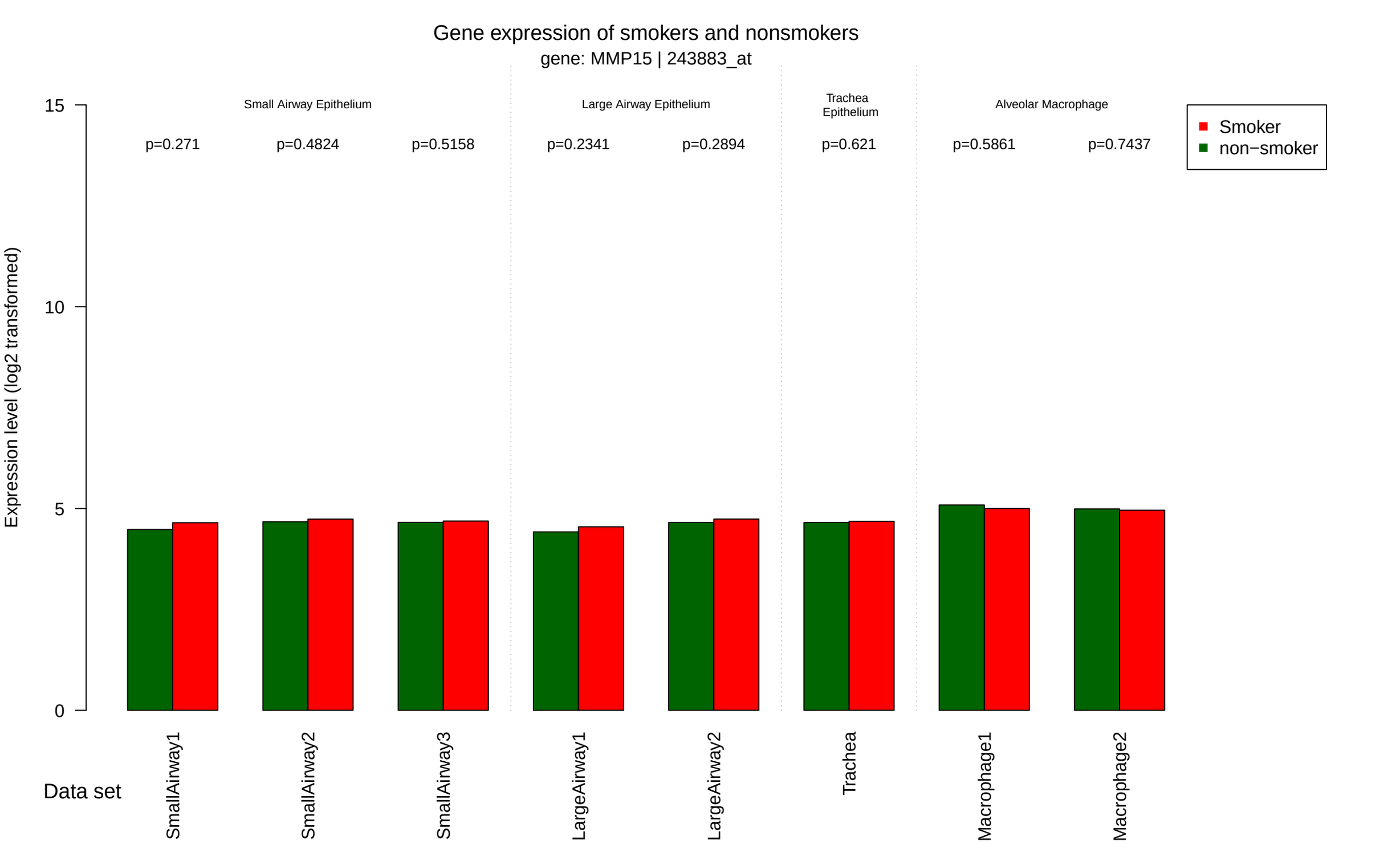
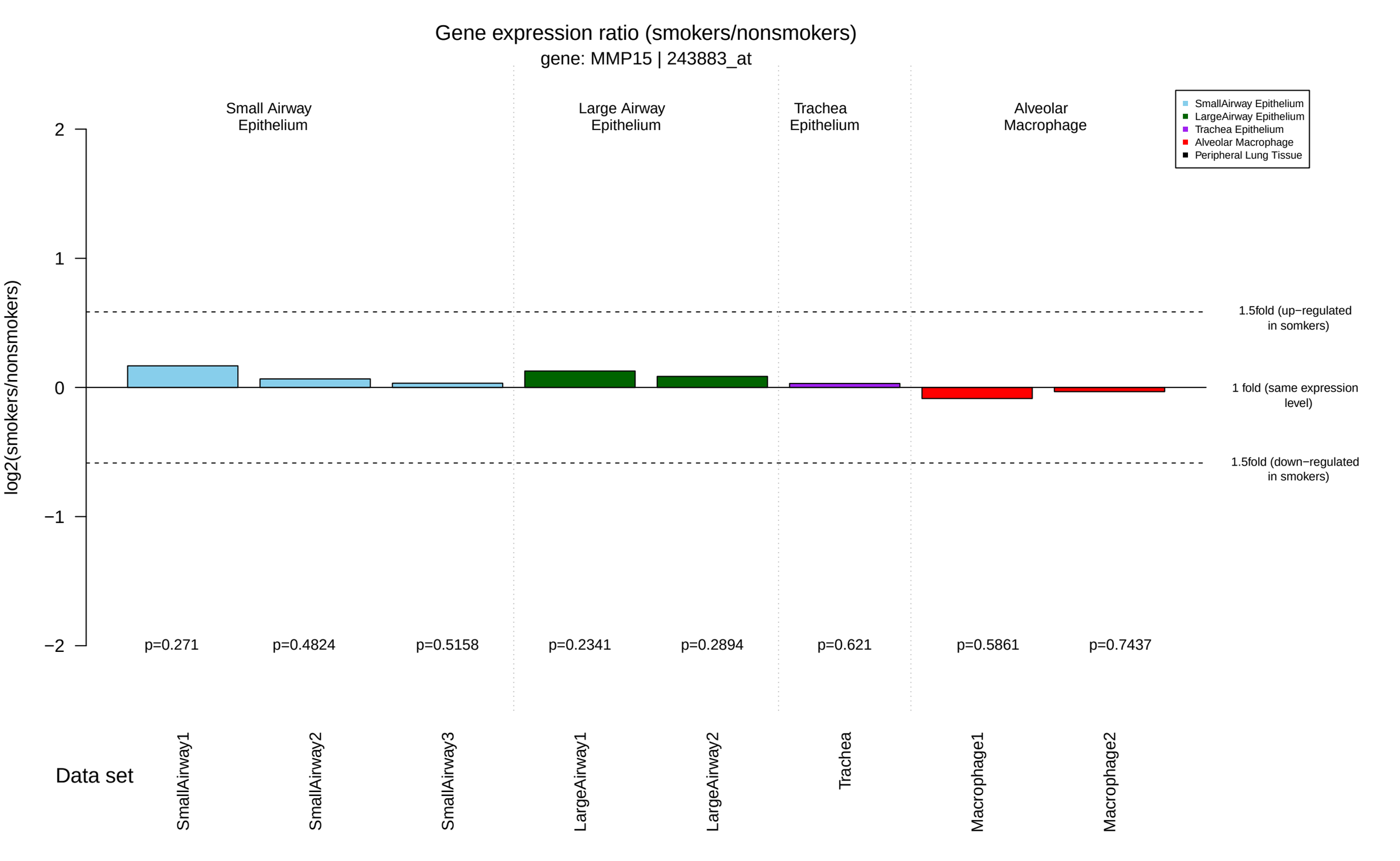
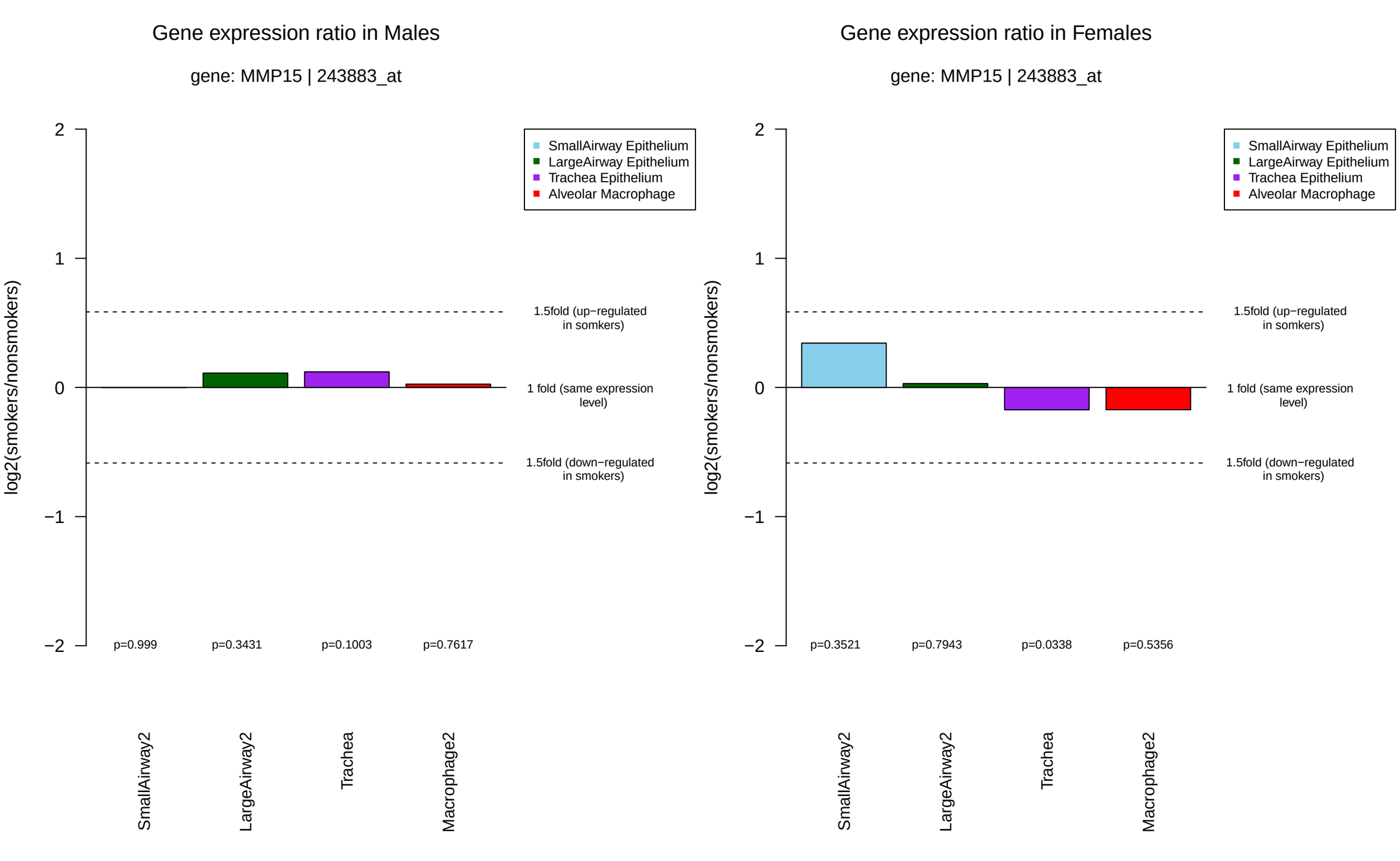
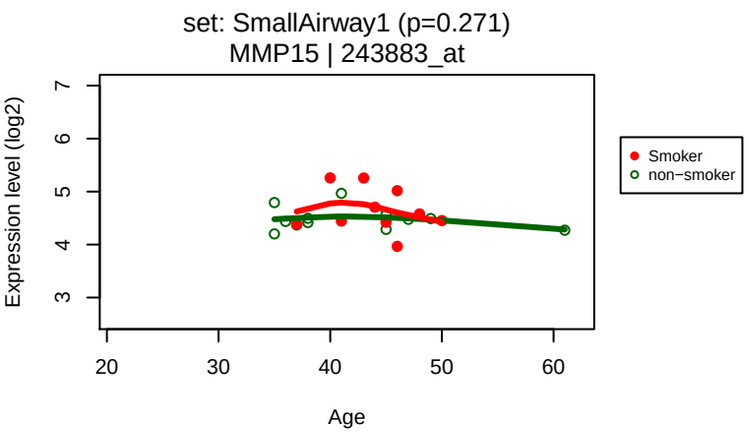
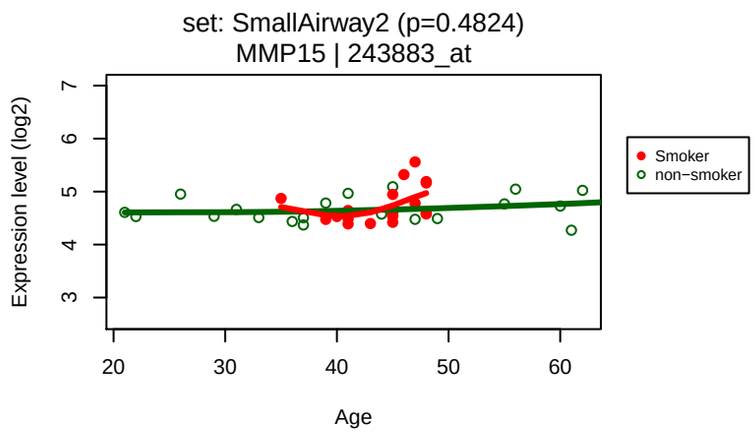
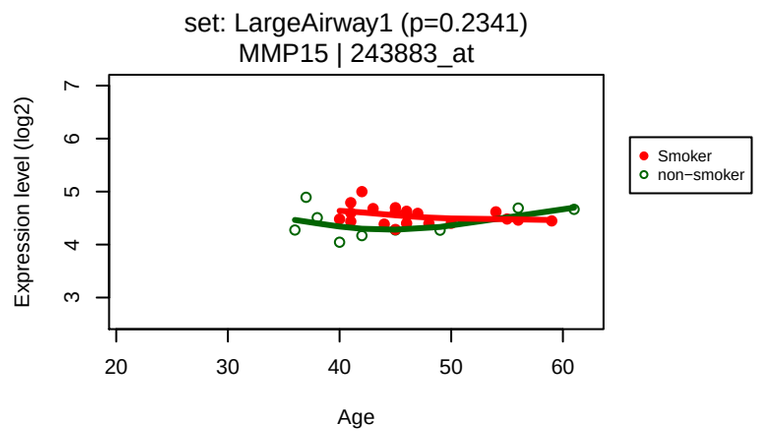

Gene expression of smokers and nonsmokers
gene: MMP15 | 243883_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.271
p=0.4824
p=0.5158
p=0.2341
p=0.2894
p=0.621
p=0.5861
p=0.7437
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

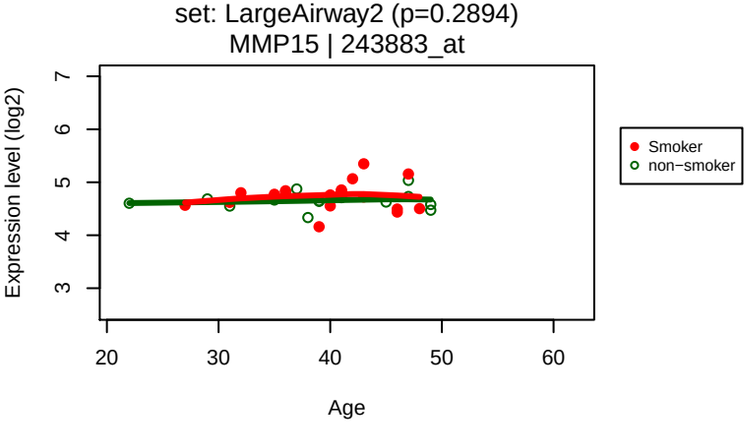

Gene expression ratio (smokers/nonsmokers)
gene: MMP15 | 243883_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.271
p=0.4824
p=0.5158
p=0.2341
p=0.2894
p=0.621
p=0.5861
p=0.7437
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

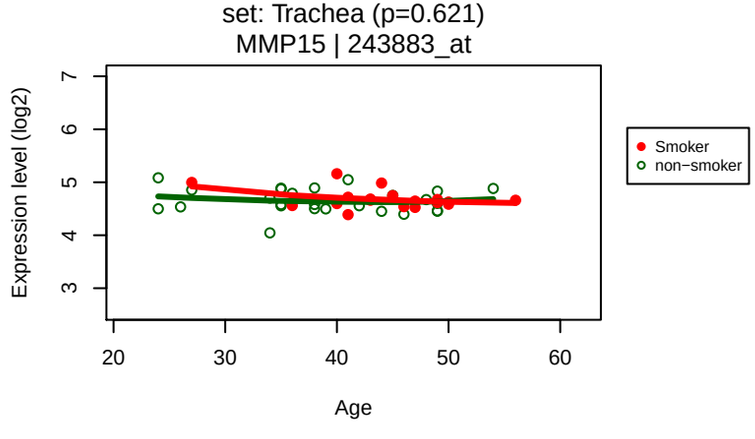

Gene expression ratio in Males
gene: MMP15 | 243883_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.999
p=0.3431
p=0.1003
p=0.7617
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP15 | 243883_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3521
p=0.7943
p=0.0338
p=0.5356
SmallAirway2
LargeAirway2
Trachea
Macrophage2

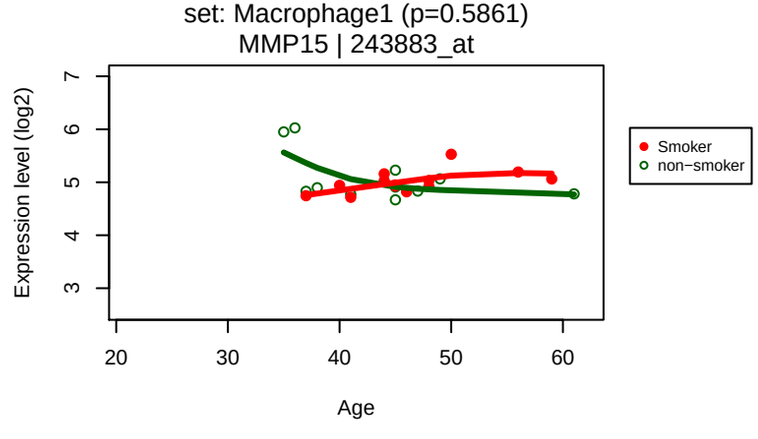

set: SmallAirway1 (p=0.271)
MMP15 | 243883_at
Expression level (log2)
Age
Smoker
non−smoker

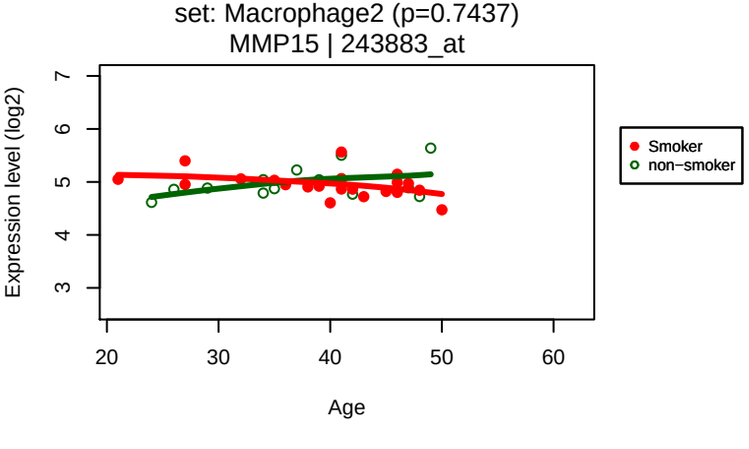

set: SmallAirway2 (p=0.4824)
MMP15 | 243883_at
Expression level (log2)
Age
Smoker
non−smoker

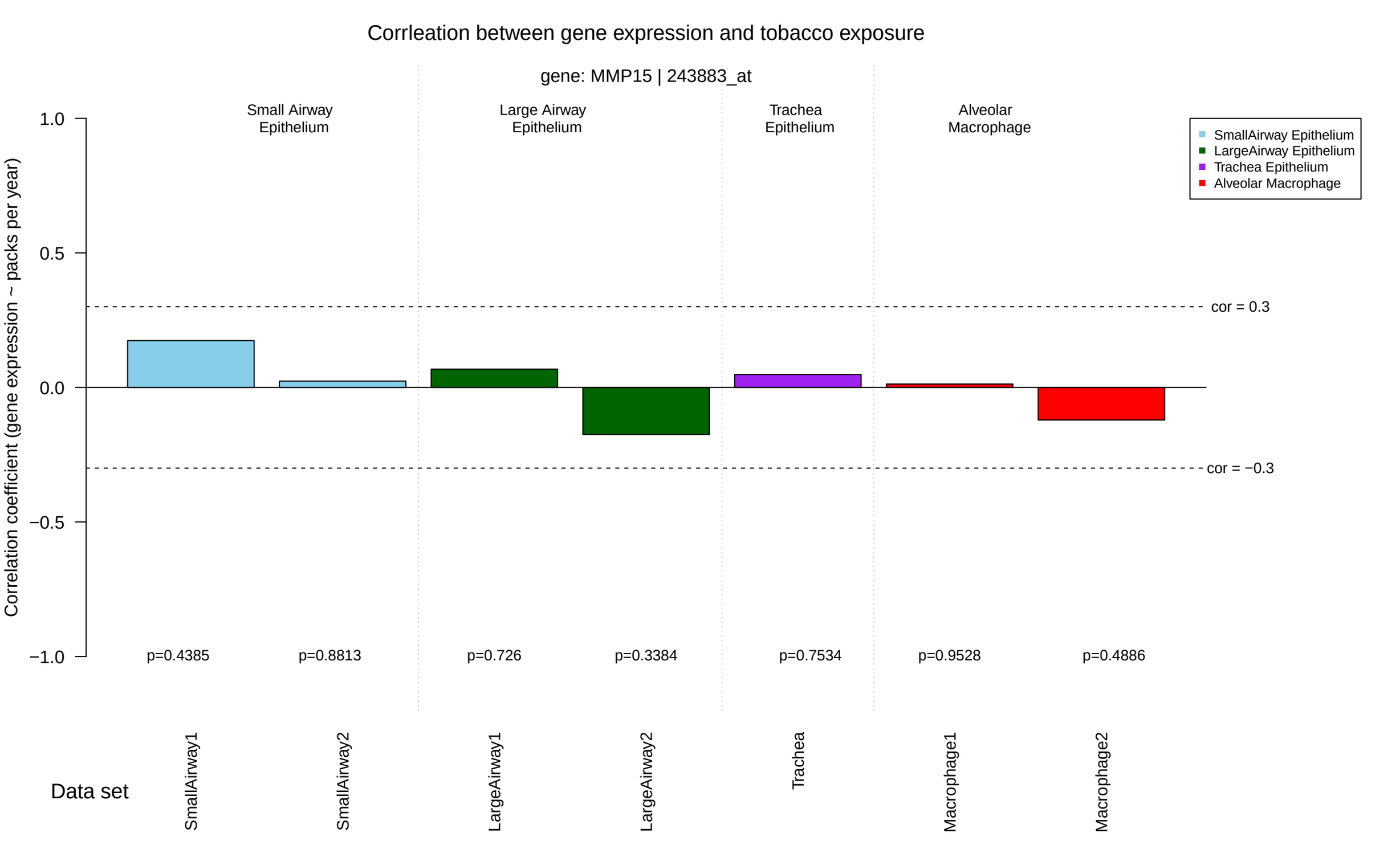

set: LargeAirway1 (p=0.2341)
MMP15 | 243883_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2894)
MMP15 | 243883_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.621)
MMP15 | 243883_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5861)
MMP15 | 243883_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7437)
MMP15 | 243883_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP15 | 243883_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4385
p=0.8813
p=0.726
p=0.3384
p=0.7534
p=0.9528
p=0.4886
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2